

Spatiotemporal modeling of COVID-19 cases using human movement and activity index[☆]

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ABSTRACT

Human interaction significantly contributes to the spread of infectious diseases like COVID-19, making it essential to consider human activities when analyzing transmission patterns. Typically, data on infectious disease transmission is aggregated across various spatial and temporal scales due to data accessibility limitations and privacy concerns. This study investigates the relationship between COVID-19 weekly incidence, human activity index, and inter-regional mobility flows at fine spatial scale.

We modeled weekly COVID-19 incidence for Middle Layer Super Output Areas in the Greater London area from March to December 2020 using Bayesian hierarchical spatiotemporal models. We constructed two types of indicators: mobility indicators by weighting travel-to-work flows with destination activity levels, and spatial indicators by weighting activities with geographic adjacency. The models accounted for area-specific spatial random effects, temporal effects, and district-level spatiotemporal effects. Both activity-based models showed modest improvements over baseline, with spatial random effects proving essential. Models using geographic adjacency performed nearly identically to those using travel-to-work estimates. Temporal stratification revealed substantial variation, with mobility-disease coefficients shifting from positive during moderate transmission to negative during explosive growth at year's end.

We presented a model to jointly study human mobility, human activity, and COVID-19 cases, providing new insights into their combined effects. Simple geographic proximity captures transmission patterns as effectively as travel-to-work data when robust spatiotemporal modeling is employed. The temporal variation in mobility-disease relationships underscores the importance of phase-stratified analyses during evolving epidemics. The models successfully reproduced spatial transmission patterns, validating their utility for surveillance applications at small areal units.

1. Introduction

1.1. Background

Understanding the transmission of infectious diseases like COVID-19 in space and time is crucial for preventing their spread. Spatial factors play a significant role in studying how infectious diseases proliferate, as infections can easily spill over into neighboring regions, with individuals potentially carrying the disease to areas they visit. Similarly, temporal aspects are important;

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the number of infected individuals on a given day typically relates to infection trends in subsequent days. One major channel of COVID-19 transmission is through anthropogenic activities. The virus spreads rapidly through the air, so if infected individuals are present in crowded places, they are likely to transmit the virus to others. Additionally, the movement of infected individuals from one region to another can introduce the disease to new areas and further propagate it within those regions. Thus, COVID-19 cases in a region can be viewed as a function of both interpersonal contact and individual movements. Reducing human activities has proven effective in controlling epidemics and pandemics; however, there has been a lack of reliable, real-time indicators to predict disease spread and guide policy decisions. This gap primarily stems from insufficient quantitative data on human activities.

Advances in location intelligence technologies now enable the analysis of digital trace data from cellular devices, offering valuable insights into human mobility and behavior for pandemic analysis and other applications. Numerous studies (Kang et al., 2020; Nepomuceno et al., 2022; Tyrovolas et al., 2021; Kraemer et al., 2020; Tokey, 2021; Slater et al., 2022) have demonstrated that human movement played a significant role in the transmission of COVID-19 using these datasets. Despite their importance, human mobility datasets necessary for understanding infectious disease transmission are difficult to obtain due to privacy concerns. Individual-level movement datasets are rare, often featuring small sample sizes that fail to capture interactions between individuals adequately, and they may not be contextualized within disease-spread scenarios. Consequently, aggregated datasets that depict mass population movements are more prevalent. For instance, Spain's daily origin–destination matrices (Ministerio de Transportes y Movilidad Sostenible (MITMS), 2024) illustrate the movement of people between regions without revealing personal details. Similarly, datasets examining travel-to-work patterns during COVID-19 in England and Wales (Office for National Statistics, 2023b), and the Advan dataset on Foot traffic for the United States of America (Advan Research, 2025), provide estimated movement patterns. Various other indices, such as Google Trends (Google, 2020) also offer insights into population movements.

1.2. Related works

The role of human mobility in the transmission of COVID-19 has been extensively studied, offering insights into how travel patterns influence the spread of infectious diseases. Kraemer et al. (2020) conducted studies linking human movement with COVID-19 spread, specifically examining Wuhan as a focal point. Their research found that early movements from Wuhan significantly influenced the spread of the virus in other Chinese provinces, although local dynamics gradually played a more pronounced role. Francetic and Munford (2021) examined commuting patterns in England from March to June 2020 using census data. Their spatial regression analysis indicated that higher commuting flows, especially those involving public transport, correlated with increased COVID-19 mortality rates. This highlights the significant role of mobility in enhancing disease transmission across regional boundaries.

Chang et al. (2021) developed a SEIR model that incorporated dynamic mobility networks from mobile phone data, capturing the spread of COVID-19 in major US cities. The model demonstrated that disadvantaged groups experienced greater risks due to crowded points of interest and less mobility reduction, highlighting socioeconomic disparities in exposure and transmission risk. Similarly, Coleman et al. (2022) employed origin–destination matrices from StreetLight Data and SafeGraph datasets to show that low-income and non-white populations in the US faced higher exposure risks, underscoring inequities in virus transmission. Meanwhile, in Denmark, Edsberg Møllgaard et al. (2022) analyzed data from mobile networks and the Facebook Data-For-Good project to evaluate mobility and social behavior during less stringent lockdowns. The results illustrated that travel restrictions did not necessarily correlate with effective containment, indicating that strategies reducing mobility must also consider various social behaviors to be successful.

Gao et al. (2021) focused on the initial outbreak phase, from January to March 2020, using metrics of urban such as Venables distance and activity density. By analyzing the time-lagged effects of human activity on COVID-19 transmission in different social and work contexts, they uncovered a lagged relationship between certain activities and the virus's reproduction number. Although their study provided valuable insights, it was limited by the subjective selection of classes and the short timeframe. The time-lag effects of mobility were also explored by Xi et al. (2020) in China, who found a 10-day lag between mobility changes and COVID-19 infections. This lag was shorter in cities with denser populations from Wuhan and more robust urban infrastructures. Their findings emphasized the need for compact urban areas to implement strict mobility restrictions, while economically wealthy cities should enforce additional non-pharmaceutical interventions.

Another important dimension in understanding infectious disease transmission is age stratification, as contact patterns governing disease spread vary substantially across age groups and settings such as schools, workplaces, and households (Mistry et al., 2021). Age-structured models have proven essential for evaluating policy interventions like school closures, demonstrating that transmission risk and the impact of control measures are highly age-dependent (Bekker-Nielsen Dunbar et al., 2024). While our mobility and activity data are not age-stratified, incorporating age-specific transmission dynamics represents an important direction for future research when such data becomes available.

These studies collectively underscore the complexity of using human mobility as an indicator for disease transmission. They highlight the necessity of employing multifaceted approaches that consider both spatial and temporal aspects of human movement to develop effective public health policies. While bringing cases into new areas is well-known, understanding how the disease spreads locally is still difficult due to smaller scales and a lack of data. For example, Slater et al. (2022) utilized movement data to redefine neighborhood structures, incorporating mobility into their models to enhance predictive performance. Although the method incorporates inter-region transmission, transmission within these regions is not considered on a smaller scale. Human activity index dataset (Mapbox, 2024) can be important for understanding these local transmission patterns by measuring potential contact possibilities.

Building on this understanding, we aim to model COVID-19 case dynamics through a combination of mobility and activity indices. We also account for the delayed effects of these factors over time. This paper contributes a novel approach to incorporating inter-regional human mobility and activity measures, known as the activity index, into spatio-temporal models. By doing so, we aim to enhance predictive performance and offer deeper insights into the interplay between mobility, activity, and disease spread.

2. Data and study area

2.1. Study area and COVID-19 cases

London, the capital and largest city of the United Kingdom is a historic and vibrant metropolis renowned for its bustling tourism and status as a major global business center. The city has a population of approximately 8.9 million people and covers an area of 1572 square kilometer (Office for National Statistics, 2024). It consists of 33 local government districts: 32 London boroughs and the City of London. Among these, twelve are designated as Inner London boroughs, while the remaining twenty comprise the Outer London boroughs. Within these districts lie smaller administrative units known as Middle Layer Super Output Areas (MSOAs), totaling 983. Our analysis focuses on these MSOAs, with a mean population of 9100 individuals.

The study utilized COVID-19 infection data for individual MSOAs across a 42-week observation window (7 March 2020 to 29 December 2020), sourced from the UK Health Security Agency's dashboard (UKHSA)(<https://ukhsa-dashboard.data.gov.uk/>). Each weekly count aggregates all individuals testing positive within seven consecutive days. A key consideration is the expansion of testing capacity during our observation period. The first 17 weeks captured only Pillar 1 testing: laboratory-confirmed cases from Public Health England facilities and National Health Service hospitals, limited to symptomatic individuals requiring clinical assessment and frontline health workers. Beginning 2 July 2020, the data incorporated Pillar 2 testing results, which broadened coverage to include the general population through diverse testing channels—local and regional testing centers, mobile facilities, and at-home test kits (UK Government, 2020).

The first COVID-19 cases in London were confirmed on February 12, 2020. The city quickly became the epicenter of COVID-19 spread within the UK during the initial transmission phase (Ghosh et al., 2020). The first lockdown was officially implemented on March 23rd, initiating extensive measures to control the pandemic's spread. This period saw a surge in infections until the end of April, after which case numbers began to decline. As illustrated in Fig. 1, several regions, including Hammersmith & Fulham, Wandsworth and Kensington & Chelsea, experienced notably high case numbers during this initial lockdown. By early May, as the number of COVID-19 cases decreased, restrictions were gradually relaxed. In the first easing phase, individuals unable to work from home were encouraged to return to their workplaces, though advised to avoid public transport. The second phase, commencing on June 1st, brought the reopening of schools and non-essential shops, alongside some relaxation of social distancing measures. The third phase, which began on July 4th, allowed pubs, restaurants, indoor theaters, and bowling alleys to reopen. These three phases are collectively referred in this paper as "Easing phases"

However, by early September, cases began to rise again, prompting the introduction of new social distancing rules on September 14th, referred to in this paper as "Voluntary restrictions". This phase included the "rule of six", which restricted gatherings to groups of six or fewer from separate households, a return to working from home, and curfews in hospital zones. During this phase, regions such as Haringey, Hammersmith & Fulham, and Hackney experienced higher incidence. The resurgence of cases eventually necessitated a second phase of stringent measures, commencing on November 5th. A marked increase in cases was observed after mid-December.

One important observation in the temporal plots is the presence of sharp fluctuations and frequent drops to zero. It is important to note that, to protect privacy, weekly counts less than 3 are censored by the UKHSA. In our dataset, these suppressed figures are recorded as zero, creating an artificial 'floor' in the data. This is confirmed by the supplementary Fig. A.8, as there are visible gap between zero and three cases. Therefore, this visible trend is likely due to this flooring, although it is possible that other minor reporting artifacts also contribute to the fluctuations.

As depicted in Fig. 2, different spatial distributions of COVID-19 cases are evident across various phases of the pandemic. These cumulative case maps highlight regional hotspots, particularly during the easing phases between June and July, where non-central MSOAs exhibited some hotspots with relatively smaller numbers. During the "Voluntary restrictions" phase, areas such as Haringey, Hammersmith & Fulham, Hackney, and several western MSOAs showed higher incidence. However, in the second lockdown, the MSOAs in the North eastern side of London within Redbridge and Havering displayed higher incidence.

2.2. Human movement and activity data

2.2.1. Activity data

Mapbox Movement data provides a de-identified and aggregated overview of mobile device activity, as outlined by Mapbox (2024). This dataset captures significant driving and non-driving activities of mobile devices, aggregated into geographic tiles (grid cells) with a resolution of approximately 100 m, forming the activity index. The collection includes location data from smart devices using the Mapbox software development kit (SDK), sourced through mechanisms such as GPS, cell tower triangulation, or router indexation when devices connect to wireless networks. Applications that use or have used the Mapbox SDK include AccuWeather, TIER, AllTrails, various car navigation systems (Rzeszewski, 2023), Snapchat and Facebook. The activity index is derived from aggregating anonymized location data into five-minute intervals within these grid cells. A high activity index within a grid-cell indicates either a high density of devices or continuous movement of the same devices—the more users located in a tile at a time,

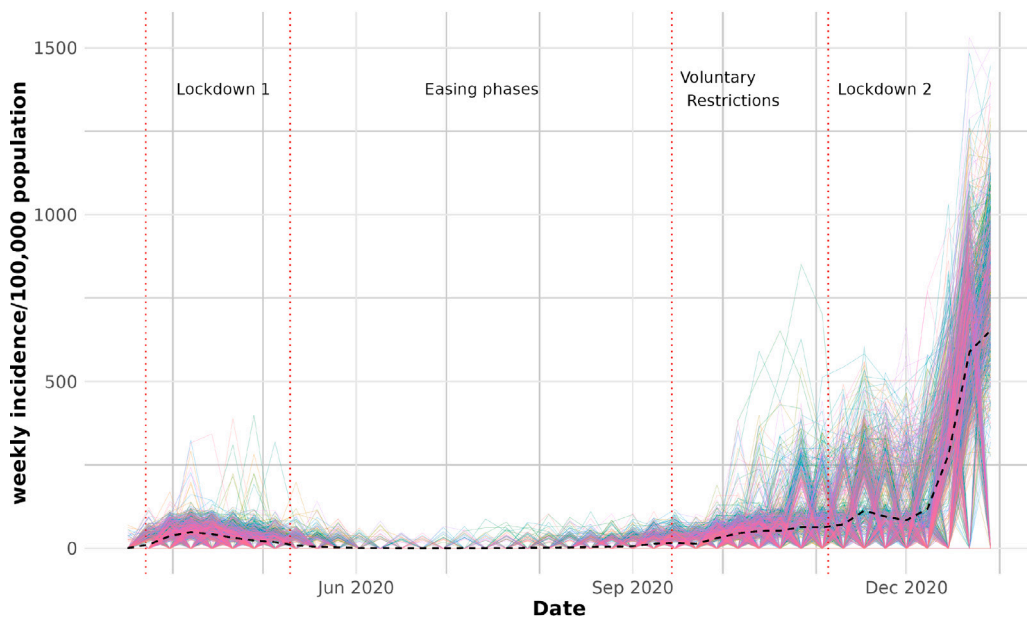


Fig. 1. Weekly case rate of COVID-19 per 100,000 inhabitants in the 983 MSOAs of Greater London area from March 2020 till December 2020. The black dashed line shows the mean case rate per week.

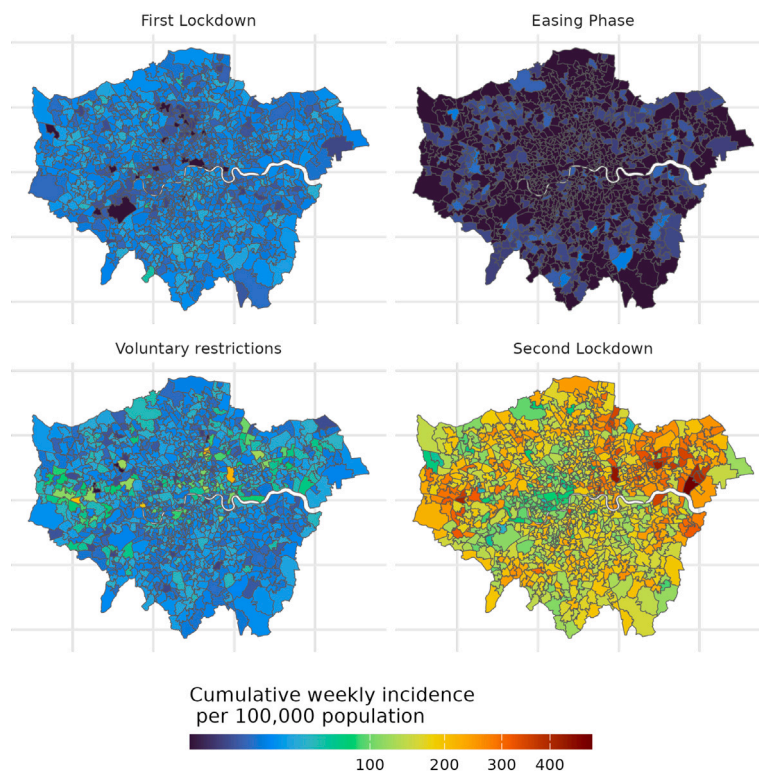


Fig. 2. MSOA wise cumulative weekly incidence per 100000 population at the end of the first lockdown (top left, 2020-03-23 to 2020-05-10), end of the easing phases (top right, 2020-06-01 to 2020-07-04), end of the Voluntary restrictions (bottom left, 2020-09-14 to 2020-11-05), and at the end of the second lockdown (bottom right, 2020-11-05 to 2020-12-31).

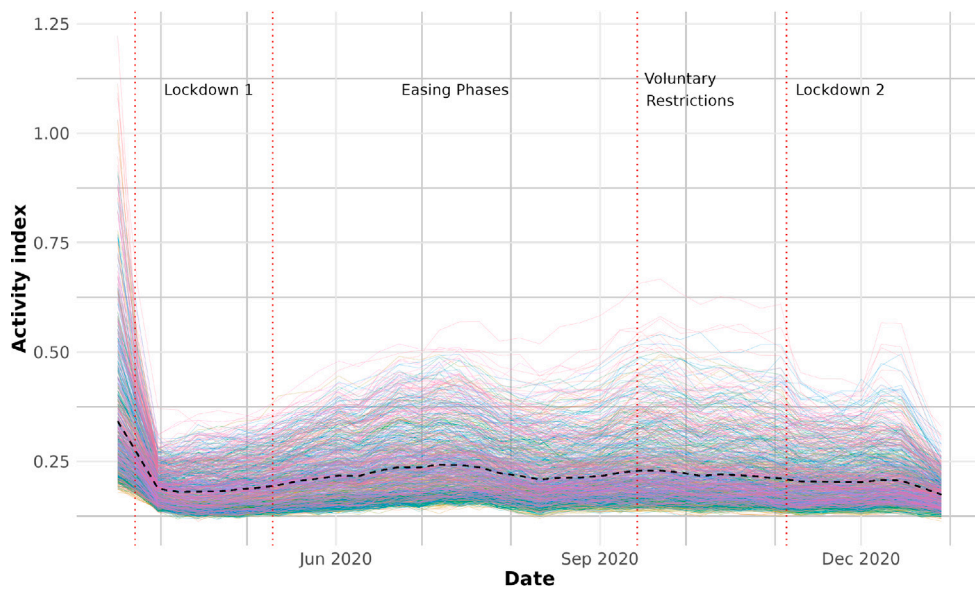


Fig. 3. Temporal series of Mapbox activity index in the 983 MSOAs of Greater London area from March 2020 till December 2020. The black dashed line shows the weekly mean activity of the study area.

the higher the human activity. Extensive calibration steps are also taken to ensure that real-world conditions are accurately reflected, accounting for fluctuations in telemetry data volumes and varied usage patterns across thousands of applications. More information on these calibration methods can be found at <https://www.mapbox.com/blog/calibrating-mapbox-movement>. To enhance privacy, Mapbox omits trip start and end points and adds a small random noise to prevent that individuals can be tracked. Additionally, tiles with only a small number of users are reported without any activity data. For further anonymization, Mapbox discards activity levels below a certain threshold and continues to add unspecified noise, applying a machine-learning algorithm to prevent re-identification.

The activity index is normalized relative to the mean activity patterns of January 2020 within the United Kingdom, rendering it unitless. This normalization establishes a country-wide baseline where an index value of 1.0 represents activity levels equivalent to pre-pandemic conditions in January 2020. Values below 1.0 indicate reduced activity compared to this baseline, while values above 1.0 indicate increased activity. This normalization approach makes the index particularly suitable for relative comparisons within the same country over short time scales, as emphasized by Mapbox, which aligns well with our analysis of 983 MSOAs in Greater London over the March–December 2020 period. Consequently, Mapbox emphasizes that this index should be used for high-level insights into human mobility trends rather than detailed individual analysis, partly due to privacy measures and data processing techniques.

The reliability of the Mapbox Movement data has been validated by the data provider through benchmarking against The Economist Normalcy Index, a comprehensive metric consisting of 8 real-world indicators across travel and mobility, commercial revenue, and retail and work activities. For the United Kingdom, the activity index demonstrates a strong correlation (Pearson $r = 0.84$) with the Normalcy Index, outperforming other publicly available mobility datasets such as Google and Apple mobility data (Mapbox, 2022). This validation, combined with the clear correspondence between observed temporal patterns in our data and known policy interventions (as discussed below and illustrated in Fig. 3), supports the use of the activity index as a reliable proxy for human mobility patterns in our spatiotemporal analysis. Detailed guidance on the calculation of the activity index is available at <https://docs.mapbox.com/data/movement/guides/>.

The activity index for the year 2020 is freely provided by Mapbox, covering data from March 16 to December 28 for the Greater London area. These activity indexes are aggregated daily, representing average activity across $100\text{ m} \times 100\text{ m}$ grids. Since the weekly COVID-19 datasets are available at the MSA level, the activity index is averaged to the same spatial and temporal resolution. Fig. 3 illustrates the trend of activity in the MSAs. A sharp decrease is evident across all MSAs following the announcement of the lockdown. As restrictions eased, activity increased, with some periods of particularly high activity in certain MSAs. After the second lockdown was announced, a noticeable decline in activity occurred again, except for a spike during the Christmas period, especially in the MSAs located in Westminster and Kensington & Chelsea. Please refer to the Tables A.5, and A.6 for the list of MSAs with high activity index in different phases.

2.2.2. Travel to work origin–destination matrices

The travel-to-work estimates developed by the UK Office for National Statistics (ONS) were utilized in this study to understand commuting behaviors during the COVID-19 pandemic (Office for National Statistics, 2023a). This dataset provides an estimate of the number of people traveling between Middle Layer Super Output Areas (MSOAs) for work during this period. The origin–destination

matrix identifies the origin as the residence and the destination as the workplace, with each volume representing the number of individuals commuting from their residential MSOA to their work MSOA. These travel-to-work matrices for England and Wales were derived using a doubly constrained gravity model. This model incorporated a population-weighted travel distance matrix, 2011 Census travel-to-work metrics, 2021 Census employment data, the National Travel Survey, and the National Trip End Model. Further details on the estimation methodology are found in [Office for National Statistics \(2023b\)](#).

The primary purpose of using the origin–destination matrices is to explore the connectivity between MSOAs in terms of the commuting population while considering the potential for disease transmission between these areas. Visualizations of the dataset, as seen in [Fig. A.6](#), indicate that central MSOAs exhibit higher volumes of incoming mobility from peripheral regions. This is likely because numerous workplaces are concentrated in central London, particularly in areas like the City of London, Westminster, Tower Hamlets, and Camden. Conversely, regions with the highest outward commuter flows include MSOAs in Newham, Tower Hamlets, and Wandsworth, primarily directed towards central areas that experience significant inbound movement. Meanwhile, the largest proportion of residents working within their own MSOAs are located in the City of London, Haringey, Tower Hamlets, Sutton, and Westminster. For detailed lists of MSOAs with significant movements, please refer to [Tables A.3](#), and [A.4](#).

3. Methods

3.1. Model

We specified a spatiotemporal Bayesian hierarchical model. The response consisted of weekly case counts of reported COVID-19 cases in the 983 MSOAs of the Greater London Area from March 16, 2020, to December 28, 2020. A Poisson distribution was assumed, here Y_{it} is the observed cases of COVID-19 in the i th MSOA of the Greater London Area on the week t ,

$$Y_{it} \sim \text{Poisson}(E_{it} \cdot \lambda_{it}), \quad (1)$$

where E_{it} is the expected number of cases in MSOA i at week t based on the population N_i , serving as an offset to account for the population at risk. The relative risk λ_{it} is modeled as:

$$\log(\lambda_{it}) = \beta_0 + \mathbf{x}_{it}^T \boldsymbol{\beta} + u_i + v_t + \eta_{jt} \quad (2)$$

where β_0 is the intercept, $\mathbf{x}_{it}$ is a vector of covariates (human mobility and activity) for the i th observation on the t th week, $\boldsymbol{\beta}$ is a vector of coefficients associated with the covariate, u_i represents the spatial random effect, modeled using a BYM ([Besag et al., 1991](#)) modified by [Riebler et al. \(2016\)](#) known as scaled BYM, v_t is the temporal random effect modeled using Random Walk of order 2, and η_{jt} is the spatiotemporal random effect modeled using type III space–time interaction ([Knorr-Held, 2000](#)) a coarser spatial scale which is Local Authority Districts (LADs) indexed in the model as j . Spatial, temporal, and spatiotemporal random effects were included to account for unobserved and unmeasured sources of variation and spatial and temporal dependency structures.

A random walk of order 2 (RW2) which refers to v_t in our model, is an extension of the basic random walk model that accounts for dependence not only on the previous time point but also on the second previous time point. This allows smoother processes and is suitable for modeling temporal trends with more gradual changes. The RW2 process captures both smooth trends and local fluctuations over time, providing a flexible framework for modeling complex temporal dynamics ([Lindgren et al., 2011](#)).

The BYM model uses spatial (ϕ_i) and non-spatial (θ_i) error terms to account for over-dispersion that is not modeled by the Poisson variates ([Besag et al., 1991](#)). When the spatial structure of the data does not fully explain the observed variance, the independent error terms will account for the rest. The spatially structured component (ϕ_i) is modeled using an Intrinsic Conditional Autoregressive (ICAR) model. This model assumes spatial dependence based on geographic adjacency by modeling (ϕ_i) as:

$$\phi_i | \phi_{-i}, \tau_\phi \sim N\left(\frac{1}{n_i} \sum_{j \sim i} \phi_j, \frac{1}{n_i \tau_\phi}\right) \quad (3)$$

where ($j \sim i$) indicates that areas (i) and (j) are neighbors, (n_i) is the number of neighbors for the area (i), and (τ_ϕ) is the precision parameter (inverse of the variance), controlling the strength of the spatial correlation. The ICAR prior is an improper prior requiring constraints (e.g., sum-to-zero) for identifiability because the precision matrix derived from the neighborhood structure is singular.

We applied the scaled version of the BYM by [Riebler et al. \(2016\)](#) which is reparameterized using penalized complexity such that the parameters have a clear interpretation also allowing for the assignment of sensible hyperparameters. In this model, the spatial effect is modeled as $u_i = (\sqrt{\rho/s})\phi_i^*(\sqrt{1-\rho})\theta_i^*\sigma$. Here, a single precision parameter σ , a mixing parameter, and scale factor s are introduced to the standard ICAR model. The mixing parameter (ρ) denotes the proportion of the variance attributed to the structured effect. The scaling factor ensures $\text{Var}(\phi_i) \approx \text{Var}(\theta_i) \approx 1$ thus by constraining the overall variance, the model avoids the need for arbitrary precision hyperpriors for (ϕ_i) and (θ_i) separately.

Similarly, we also introduce a spatio-temporal effect, refers to η_{jt} in our model, at 33 LADs within the Greater London area. Please refer to [Fig. A.7](#) to view the map depicting the LAD in the study area. We introduced a LAD-specific temporal effect to capture variations in weekly COVID-19 incidence across the 33 LAD in the Greater London Area. This was achieved using a Random Walk of order 2 (RW2) model. The RW2 model assumes that the value at each time point depends on two previous time point, allowing for the modeling of trends that may vary bi-weekly within each LAD.

Table 1
Model performance comparison. Lower values indicate better model fit.

SN	Model	WAIC	CPO	RMSE
<i>Models with Spatial Random Effects</i>				
1	Baseline	1,369,496	22.38	75.96
2	Neighbor Activity	1,369,345	22.38	75.91
3	Mobility Activity	1,369,774	22.37	75.91
<i>Reduced Models without Spatial Random Effects</i>				
4	Reduced Neighbor Activity	1,254,727	29.55	95.45
5	Reduced Mobility Activity	1,254,520	29.55	95.47

3.2. Defining movement indicators

The travel-to-work origin–destination (OD) matrix which estimates the number of people moving between regions was combined with the activity index within those areas to form a combined movement indicator. Regions with high activity levels are presumed to facilitate increased contact between infected and non-infected individuals, thus contributing to disease spread and the estimated movements between regions suggest that these contacts are established through people traveling from other regions.

An analysis of the MSOAs with high mean activity index reveals that a few consistently exhibit a very high activity index throughout different phases of COVID-19 pandemic (refer to the [Tables A.5](#), and [A.6](#)). It indicates that these regions are highly active regions with constant presence of people. Similarly, the exploration of the travel-to-work OD matrix suggests that the flow is high into those selected regions ([Tables A.3](#), [A.4](#)). Thus, instead of directly correlating this activity index with COVID-19 infection rates, we combine them with the origin–destination matrices by creating the mobility indicator to understand their combined effect better, as defined in subsequent paragraphs.

Let M be the origin–destination (OD) matrix where m_{ij} denotes the number of people commuting from region i to region j , and $A_t = \{a_{it}\}^\top$ as the vector of mean daily activities for each region i during week t . The diagonal of M is set to zero ($\text{diag}(M) := 0$) to exclude intra-region commuting, as within-region travel-to-work counts are incorporated as a separate covariate in the model.

For the entire study period spanning T weeks, we construct a block-diagonal matrix to compute the mobility indicator vector I as:

$$I = (I_M \otimes M) A \quad (4)$$

where I_M is a $T \times T$ identity matrix, $\otimes$ denotes the Kronecker product, and $A = [A_1^\top, A_2^\top, \dots, A_T^\top]^\top$ is the stacked vector of activities across all weeks. This yields the time-specific indicator vectors:

$$I_t = M A_t, \quad t = 1, 2, \dots, T \quad (5)$$

For region i in week t , the mobility indicator $I_t[i] = \sum_{j \neq i} m_{ij} a_{jt}$ represents a weighted sum of outgoing commuters to destination region j , weighted by the activity level a_{jt} in those destinations. This formulation captures transmission risk when commuters (traveling to work) return home after visiting regions with varying activity levels.

Spatial activity indicators. In addition to mobility-based indicators derived from commuting data, we construct spatial activity indicators using geographic proximity as an alternative proxy for spatial coupling. We compute the spatial indicator as:

$$I_t^{(W)} = W A_t \quad (6)$$

where W is a row-standardized spatial weight matrix constructed from the first-order queen contiguity neighborhood structure of the 983 MSOAs. Two regions are considered neighbors if they share a common border or vertex. For each region i with n_i neighbors, the spatial weights are defined as:

$$w_{ij} = \begin{cases} \frac{1}{n_i} & \text{if regions } i \text{ and } j \text{ are adjacent} \\ 0 & \text{otherwise} \end{cases} \quad (7)$$

This row-standardization ensures each row of W sums to unity, giving equal weight to all neighbors of region i regardless of the total number of neighbors. Unlike the mobility indicator $I_t^{(M)}$, which incorporates directional commuting flows weighted by destination activity, the spatial indicator $I_t^{(W)}$ represents a spatially-lagged average of neighboring regions' activities. This allows us to assess whether actual travel patterns or simple geographic adjacency better explain COVID-19 transmission dynamics.

The mobility indicator $I_t^{(M)}$ captures transmission risk through actual human movement between regions, weighted by activity levels at destinations. In contrast, the spatial indicator $I_t^{(W)}$ represents a simpler hypothesis: that transmission is driven by spatially-lagged activity in neighboring regions regardless of commuting patterns. Comparing these two formulations helps determine whether data on commuting for work provides explanatory power beyond basic geographic adjacency.

3.3. Model fitting and evaluation

We compared a total of five hierarchical spatiotemporal models to evaluate the impact of incorporating mobility and activity data on model performance. The first three models share a common baseline structure with: (i) a spatial random effect u_i using scaled BYM priors to capture spatial correlation between MSOAs, (ii) a temporal random effect v_t modeled through a second-order random walk (RW2) to capture weekly temporal trends, and (iii) a spatiotemporal random effect η_{it} at the Local Authority District (LAD) level, also modeled using RW2, to account for coarser-scale spatiotemporal variation across the 33 LADs in Greater London. The three models differ in their covariate specifications:

Model 1 (Baseline): Contains only the random effects structure described above, with no mobility or activity covariates.

Model 2 (Neighbor Activity Model): Extends the baseline by adding three types of covariates: (i) within-region activity index, (ii) within-region travel-to-work movement, (iii) the spatial indicator ($I_i^{(W)}$) that incorporates spatially weighted activity from neighboring MSOAs.

Model 3 (Mobility Activity Model): Also extends the baseline with within-region activity and travel-to-work movement, but replaces the spatial indicator ($I_i^{(W)}$) with the mobility indicator ($I_i^{(M)}$) that weights inter-MSOA activity by the travel-to-work origin–destination flows rather than by geographic adjacency.

To assess spatial confounding, we fit sensitivity versions of Models 2 and 3 by removing the BYM2 spatial random effect u_i while retaining all covariates and temporal structure. These reduced models — the **Reduced Neighbor Activity Model** (Model 4) and **Reduced Mobility Activity Model** (Model 5) — allow comparison of coefficient estimates with and without spatial structure, diagnosing whether spatial random effects absorb or distort spatially-patterned covariate effects.

Model parameters of the above mentioned models were estimated in the Bayesian framework, using Integrated Nested Laplace Approximation (INLA) and the R package R-INLA (Rue et al., 2009). Parameter uncertainty was accounted for by assigning prior distributions to the parameters. A penalized complexity prior (Simpson et al., 2017) was used as the prior distribution on the hyperparameters of spatial, temporal and spatiotemporal random effects with scale parameter (σ_0) equal to 0.25 and α equal to 0.01. Similarly, default non-informative priors from the package R-INLA were used for the other parameters. Here, the intercept of the model is assigned a Gaussian prior with mean and precision equal to 0. The parameters for the covariates are assigned Gaussian priors with mean equal to 0 and precision equal to 0.001.

We estimated the goodness of the fit measure to select the best-fitting model using the Watanabe Akaike information criterion (WAIC), which is an information criteria measure that estimates goodness of fit but also penalizes on the number of parameters or complexity of model (Watanabe and Opper, 2010). Similarly, we also computed the conditional predictive ordinate (CPO) of the models (Pettit, 1990). CPO expresses the posterior probability of observing the value (or set of values) of y_i when the model is fitted to all data except y_i . We summarized the CPO by calculating the cross-validated (CV) log score (Gneiting and Raftery, 2007) to compare the models. Along with these measures, the mean squared error of each model was calculated to compare model fits.

3.4. Computational implementation

All models were implemented in R (version 4.3.2) using the INLA package (version 24.02.09) (Rue et al., 2009), which utilizes parallel processing by default. Computations were performed on a Lenovo ThinkPad P14s Gen 4 workstation with an Intel Core i7-13th generation processor and 64 GB RAM. For our dataset of 983 MSOAs over 42 weeks (41,286 observations), computational time ranged from approximately 12 s for baseline models to 13 s for full spatiotemporal models with indicators and other covariates. INLA utilized up to 32 threads with an average efficiency of approximately 27%.

4. Results

We analyzed COVID-19 weekly incidence across 983 London MSOAs over 42 weeks (March 2, 2020 to December 31, 2020), comparing the five spatiotemporal models, as described in section 3.3 to assess the value of incorporating human mobility data, and activity indicators. Similarly, we performed sensitivity analysis to identify the possibility of confounding between the spatial random effect and the indicators. We focus on comparing the results from the two models that incorporate the indicators. Finally, we explore the changing direction of the effect of the indicators in the mobility activity model.

4.1. Model comparison and performance

We compared five spatiotemporal models to assess the value of incorporating human mobility data and the role of spatial random effects (Table 1). Models 1–3 include BYM2 spatial random effects; Models 4–5 are sensitivity versions excluding spatial structure.

Among spatially-structured models, the Neighbor Activity Model (Model 2) achieved the lowest WAIC (1,369,345), while both activity models showed nearly identical goodness of fit (RMSE = 75.91). Both activity models showed marginal improvements over baseline, suggesting modest added value from mobility covariates beyond spatial–temporal structure.

The reduced models (Models 4–5) demonstrate worse fit (RMSE increases from 75.91 to 95.5; CPO worsens from 22.4 to 29.6). Although the WAIC suggests improvement in the reduced models, this is driven by criterion’s penalty on the model complexity — specifically the large number of effective parameters in the BYM2 structure to fit the local variations. The improved fit with spatial random effects (improved RMSE and CPO) indicate they capture essential spatial structure — including spatial autocorrelation and

Table 2

Sensitivity of fixed effect estimates to spatial random effects. Mean estimates reported.

Covariate	(Neighbor activity)		(Mobility activity)	
	Full(2)	Reduced (4)	Full (3)	Reduced (5)
Spatial Indicators	1.90	−0.54		
Mobility Indicators			−9.49	2.59
Weekly mean activity	0.54	−3.20	0.56	−3.22
Within-region flow	0.000044	0.000024	0.00024	−0.000011

confounding from unmeasured factors like demographics, housing density, and healthcare access — that our mobility and activity covariates cannot represent.

Models 2 and 3 performed nearly identically (WAIC difference = 429; RMSE = 75.91 for both), indicating that travel-to-work origin–destination data provides minimal advantage over geographic adjacency. This likely reflects limitations in the origin–destination data: it captures only workplace commuting (excluding other travel purposes) and represents annual aggregates rather than weekly variation aligned with our activity and COVID-19 incidence data. These temporal and functional mismatches may explain why granular mobility data offers little improvement, suggesting that, in its current form, potentially expensive OD data may not be necessary for surveillance when robust spatial–temporal models are employed.

4.2. Sensitivity to spatial random effects

To assess whether spatial random effects confound the estimation of covariate effects, we conducted sensitivity analyses comparing coefficient estimates between the full models (Models 2 and 3) and their reduced counterparts without spatial structure (Models 4 and 5).

Table 2 reveals substantial coefficient instability when spatial random effects are removed. In both model pairs, activity coefficients reverse sign. Similarly, within-region weekly activity coefficients change from positive to negative in both model pairs. This demonstrates spatial confounding—the inability to separate effects of spatially-structured covariates from unmeasured spatial confounders when variation occurs primarily between rather than within regions.

This coefficient instability arises from spatial confounding. High-activity regions like Westminster differ from low-activity regions not only in activity levels but also in demographics, housing density, healthcare access, and other factors that affect COVID-19 risk. When spatial random effects are excluded, the fixed effects capture both the within-region effect of activity changes over time and stable between-region differences in baseline risk that correlate with activity patterns—making it hard to distinguish genuine temporal associations from confounding by regional characteristics.

Including spatial random effects accounts for these persistent regional differences, substantially improving goodness of fit (RMSE from 95.5 to 75.9) and allowing coefficients to better reflect within-region temporal variation. However, this creates a trade-off: spatial random effects account for regional confounding but may also remove part of the relationship between activity and disease if that relationship reflects stable differences between regions rather than changes over time within regions. Given this sensitivity, we interpret coefficients cautiously, focusing on model comparisons and temporal evolution rather than individual parameter values.

4.3. Temporal dynamics of mobility indicator

We explored the temporal evolution of the mobility indicator coefficient. The aggregate coefficient from the full study period was counterintuitive (negative), which prompted us to investigate whether this relationship changed over time. We fitted the Mobility Activity Model (Model 3) using cumulative data ending at successive two-week intervals to trace how the mobility-disease relationship evolved across epidemic phases. We began this analysis from the third week of September (one week after voluntary restriction measures were enforced) and continued through the end of the study period, with particular focus on the second wave of the pandemic.

While spatial confounding affects the absolute magnitude of coefficients, temporal patterns in how these coefficients change remain informative because the confounding structure likely remains relatively stable over time. Fig. 4 shows the evolution of the mobility indicator coefficient over the study period, revealing a substantial shift from positive to negative values as the epidemic progressed through different phases.

Fig. 4 reveals substantial temporal variation in coefficient estimates across the study period. The mobility indicator coefficient exhibits a striking shift from strongly positive (3.27, 95% CI: 1.99–4.53 in late September) to strongly negative (−9.47, 95% CI: −10.33 to −7.08 in late December). In contrast, the weekly mean activity coefficient remains positive throughout the study period, though its magnitude varies over time.

This temporal evolution characterizes three distinct epidemic phases:

Phase 1 (Voluntary restrictions: September 22–November 5, 2020): Positive mobility effect. During the voluntary restriction period when weekly incidence ranged from 15–65 per 100,000 per week, the mobility indicator showed a strong positive association with COVID-19 incidence (0.38–3.27). This aligns with theoretical expectations that inter-regional commuting drives spatial disease spread during periods of moderate transmission.

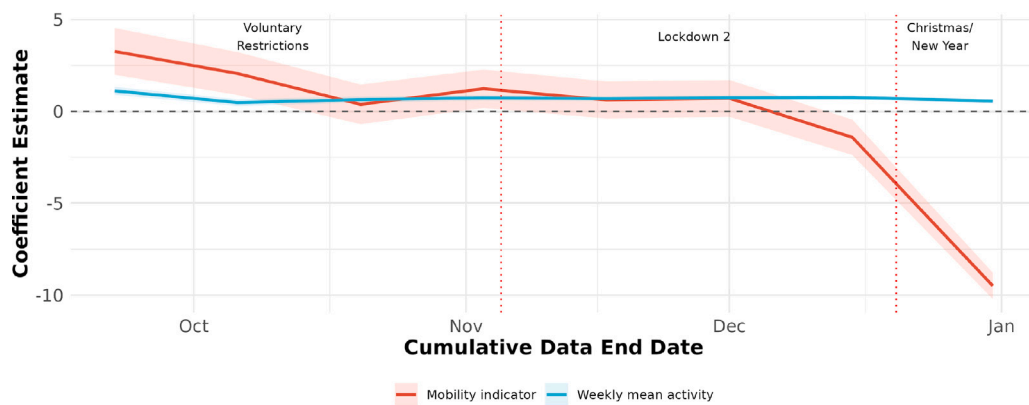


Fig. 4. Temporal evolution of the coefficients.

Phase 2 (Lockdown 2: November 5–December 20, 2020): Weakening positive effect. Following the implementation of England's second national lockdown, as weekly incidence increased to 70–280 per 100,000 per week, the mobility coefficient remained positive but declined substantially (0.6–1.23), likely reflecting behavioral adaptations, policy restrictions reducing non-essential mobility, and shifts towards essential-only travel.

Phase 3 (Christmas/New Year period: December 20, 2020 onwards): Negative association. During the Christmas and New Year period when weekly incidence exploded to around 600 per 100,000 per week, the mobility coefficient became strongly negative. This counterintuitive result likely reflects multiple confounding mechanisms rather than a protective effect of mobility.

This counterintuitive negative association likely reflects multiple confounding mechanisms. Areas with severe outbreaks experienced the greatest behavioral mobility reductions, creating spurious negative correlations through reverse causation. At very high prevalence, transmission became dominated by unavoidable household and essential service contacts, with discretionary mobility contributing less to case variation. Finally, the more transmissible Alpha variant (B.1.1.7) achieved dominance in London by late December 2020 (Davies et al., 2021), shifting transmission towards close-contact settings where larger-scale activity/mobility plays a lesser role.

The contrasting temporal patterns between the mobility indicator and weekly mean activity coefficient suggest these covariates capture different aspects of transmission. The mobility indicator, based on origin–destination flows, appears more sensitive to behavioral changes and epidemic phase transitions, while the within-region activity measure shows greater stability over time.

4.4. Spatial patterns

Fig. 5 presents a comparison between model-fitted and observed incidence across four selected weeks representing different epidemic phases. The model successfully captures the general spatial trends of infection across most periods. For the week of March 31, 2020 (Lockdown 1), the model accurately identifies regions with higher incidence, particularly in outer boroughs. During the week of August 25, 2020 (lockdown easing phase), incidence declined substantially with only a few regions experiencing elevated rates; the model performs worse during this period due to the overall low incidence and limited spatial variation.

For the week of November 10, 2020 (voluntary restrictions), when cases began rising again, the model excels by accurately pinpointing hotspots and capturing the clear west-to-east spatial gradient in transmission. During the week of December 22, 2020 (Lockdown 2), the model effectively captures the overall high transmission levels across London but produces smoother spatial patterns than observed. The spatial random effects component tends to smooth over local variation, which explains why fitted maps show more uniform patterns than observed data, particularly during periods of extreme incidence (very low or very high). Overall, the model performs well except in scenarios with generally low incidence and isolated high-case areas, where limited spatial signal makes accurate model fit challenging. Temporal series of observed and fitted incidence for randomly selected regions are included in the annex (Figs. A.9), further demonstrating model performance across time.

5. Discussion

This study examines the relationship between human mobility and COVID-19 transmission across London's 983 MSOAs during 2020. While mobility data modestly improved model fit, the Neighbor Activity Model (geographic adjacency) and Mobility Activity Model (commuting flows) performed nearly identically, suggesting that origin–destination data offers limited additional value when robust spatial-temporal models are employed. This equivalence likely reflects limitations in the OD matrix — annual aggregation and restriction to workplace commuting — rather than indicating that mobility patterns are fundamentally uninformative. The most striking finding is the temporal heterogeneity in the mobility-disease relationship: coefficients shifted from positive (3.27) during late September to negative (−9.47) towards the end of December, reflecting behavioral responses, transmission saturation favoring

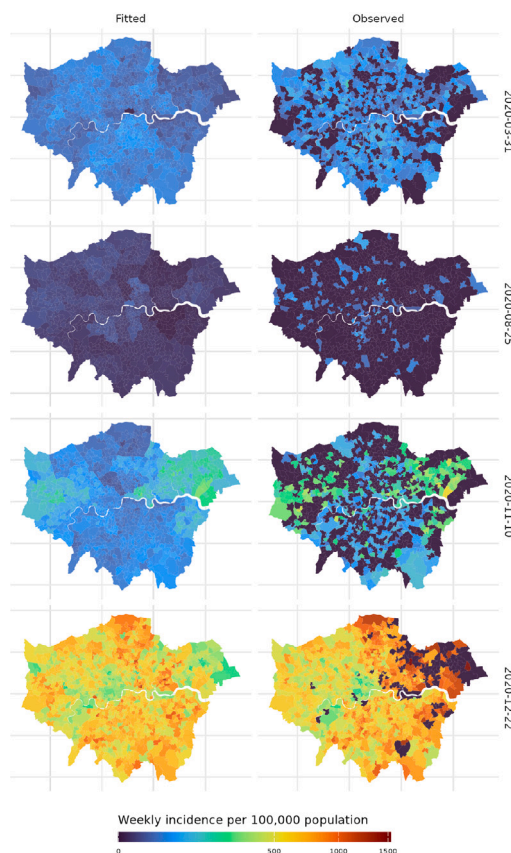


Fig. 5. Fitted and observed COVID-19 weekly incidence per 100,000 population for four selected weeks representing different pandemic phases. Left panels show model-fitted weekly incidence; right panels show observed weekly incidence.

household contacts, and Alpha variant emergence. This demonstrates that time-aggregate estimates obscure critical phase-dependent dynamics, with important implications for surveillance systems and policy assessment. Assessments of mobility interventions should focus on moderate transmission periods when relationships are clearest, rather than aggregating across fundamentally different epidemic phases.

Our sensitivity analyses revealed substantial spatial confounding — coefficient sign reversals when spatial random effects were removed — affecting both activity models. This represents a fundamental challenge in spatial analysis: spatially-structured covariates cannot be clearly separated from unmeasured spatial confounders when variation occurs primarily between rather than within regions. Including spatial random effects improved model fit substantially (RMSE from 95.5 to 75.9) by capturing persistent regional differences, but creates a trade-off where spatial effects may capture both true indicators–transmission relationship and residual spatial confounding simultaneously. This necessitates cautious interpretation, focusing on model comparisons and temporal patterns rather than individual coefficients. Our approach treated spatial coupling structures (OD matrix M , weight matrix W) as fixed, but parametrizing these weights — estimating distance-decay or flow-volume parameters from data (Meyer and Held, 2014) — could provide deeper insights.

Several limitations are worth noting. The travel-to-work OD matrix represents annual aggregates of workplace commuting, unable to capture weekly variations or behavioral changes during lockdowns, and excludes non-work travel (shopping, recreation, social visits). These limitations likely explain why detailed flow data offered minimal advantage over simple adjacency. The Mapbox activity data, while capturing human presence at fine scales, cannot measure protective behaviors (masking, distancing) or symptom status, exhibits smartphone user bias, and was available only for 2020. Our study period captured multiple epidemic waves driven by variants with distinct transmission characteristics — particularly Alpha (B.1.1.7), which achieved dominance by late December with substantially higher transmissibility — but we could not incorporate variant-specific parameters. Weekly aggregation may obscure daily-level dynamics relevant to COVID-19's short generation interval. Our analysis examined only reported cases as outcomes and activity/mobility as covariates, with numerous potentially relevant factors (demographics, socioeconomic status, healthcare capacity, testing rates) absorbed in spatial random effects. Future work with daily or weekly mobility data capturing diverse travel purposes could reveal whether dynamic patterns substantially improve model fit. Examining finer spatial scales, additional outcomes (hospitalizations, deaths), and incorporating variant-specific parameters and behavioral indicators would enable more nuanced understanding of mobility–transmission relationships. In conclusion, human mobility data modestly enhances model fit, but the

relationship exhibits marked temporal heterogeneity that complicates interpretation. Simple geographic structures perform as well as detailed flow data when robust spatial–temporal models are employed, though this likely reflects data limitations rather than fundamental mobility irrelevance. Spatial confounding and temporal complexity necessitate cautious interpretation of aggregate associations. Despite these challenges, the models successfully captured spatial transmission patterns, validating their surveillance utility. Our findings underscore the importance of temporal stratification in epidemic analyses and phase-dependent dynamics during rapidly evolving epidemics.

Appendix

See Figs. A.6–A.9 and Tables A.3–A.6.

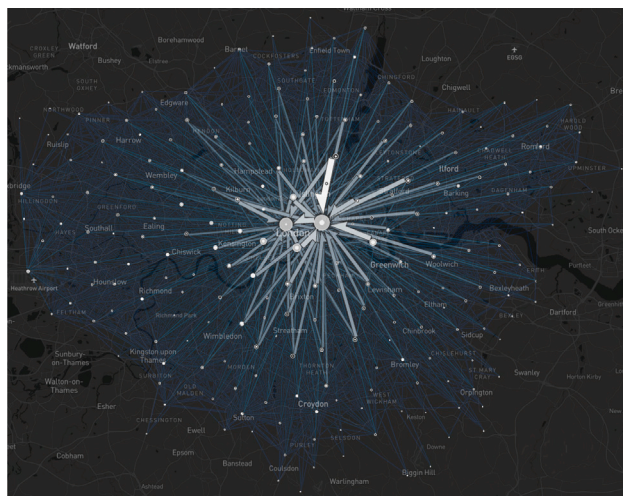


Fig. A.6. Flowmap of the travel to work origin–destination matrix. Here larger arrow size refers to larger volume of travel between the MSOAs. Prepared using R package flowmapblue.R (<https://github.com/FlowmapBlue/flowmapblue.R>)

Table A.3

Top 10 MSOAs by incoming and outgoing commuters for work.

Incoming commuters			Outgoing commuters		
SN	MSOA name	Count	SN	MSOA name	Count
1	City of London 001	264 208	1	Newham 013	8665
2	Westminster 018	113 898	2	Tower Hamlets 028	7315
3	Westminster 013	96 907	3	Newham 037	6396
4	Tower Hamlets 033	74 849	4	Wandsworth 002	6124
5	Westminster 020	64 533	5	Greenwich 036	5874
6	Camden 028	50 337	6	Barnet 030	5610
7	Southwark 002	41 049	7	Wandsworth 010	5081
8	Westminster 011	36 353	8	Lewisham 012	4935
9	Camden 026	29 662	9	Hackney 021	4928
10	Islington 022	27 386	10	Southwark 003	4918

Table A.4

Top 10 MSOAs with most commuters traveling to work within MSOAs.

SN	MSOA name	Within movement
1	City of London 001	567
2	Hillingdon 031	294
3	Tower Hamlets 033	179
4	Croydon 027	164
5	Westminster 013	157
6	Westminster 018	145
7	Hillingdon 015	135
8	Kingston upon Thames 009	129
9	Havering 013	106
10	Hillingdon 016	104

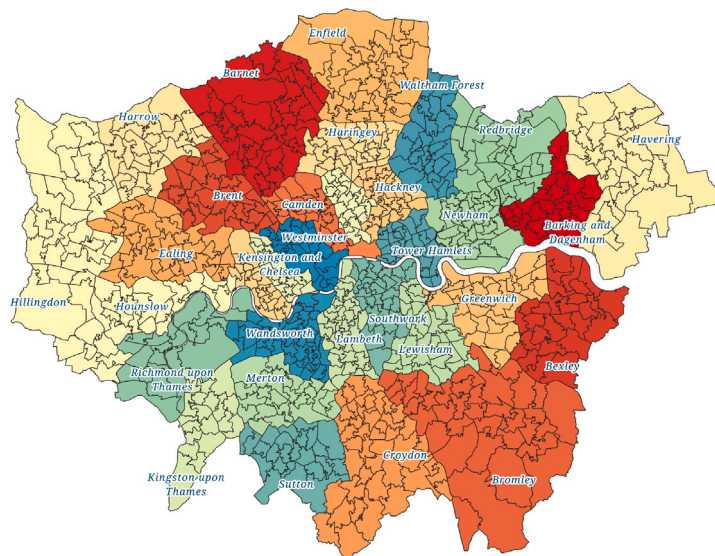


Fig. A.7. Local Authority District(LAD) and their Middle layer Super Output Area (MSOAs) of the Greater London area.

Table A.5

Top MSOAs by weekly mean activity: First lockdown and easing phase.

First lockdown				Easing phase			
Rank	Date	Mobility	MSOA name	Rank	Date	Mobility	MSOA name
1	03–23	0.646	Westminster 012	1	09–07	0.613	Westminster 013
2	03–23	0.583	Kens. & Chelsea 015	2	08–31	0.583	Westminster 013
3	03–23	0.579	Tower Hamlets 025	3	07–20	0.569	Westminster 013
4	03–23	0.559	Tower Hamlets 021	4	07–13	0.569	Westminster 013
5	03–23	0.542	Camden 025	5	08–24	0.567	Westminster 013
6	03–23	0.525	Tower Hamlets 013	6	08–17	0.558	Westminster 013
7	03–23	0.523	Hackney 02	7	09–07	0.552	Westminster 011

Table A.6

Top MSOAs by weekly mean activity: Voluntary restriction and second lockdown.

Voluntary restriction				Second lockdown			
Rank	Date	Mobility	MSOA Name	Rank	Date	Mobility	MSOA Name
1	09–21	0.667	Westminster 013	1	12–07	0.567	Westminster 013
2	09–14	0.657	Westminster 013	2	12–14	0.565	Westminster 013
3	09–28	0.635	Westminster 013	3	12–07	0.518	Westminster 011
4	10–12	0.627	Westminster 013	4	12–14	0.512	Westminster 011
5	10–19	0.619	Westminster 013	5	12–07	0.510	Westminster 012
6	11–02	0.609	Westminster 013	6	12–14	0.496	Kens. & Chelsea 014
7	10–05	0.609	Westminster 013	7	12–14	0.490	Westminster 012

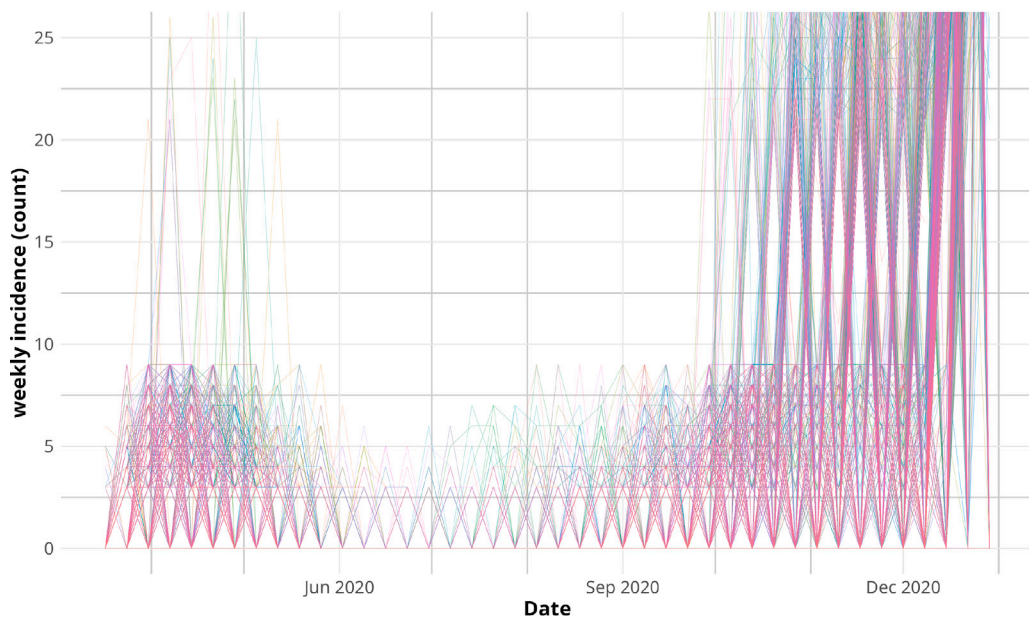


Fig. A.8. Weekly COVID-19 cases in the study area. The y-axis is truncated at 25 cases, and the gap between 0 and 3 cases reflect privacy censoring.

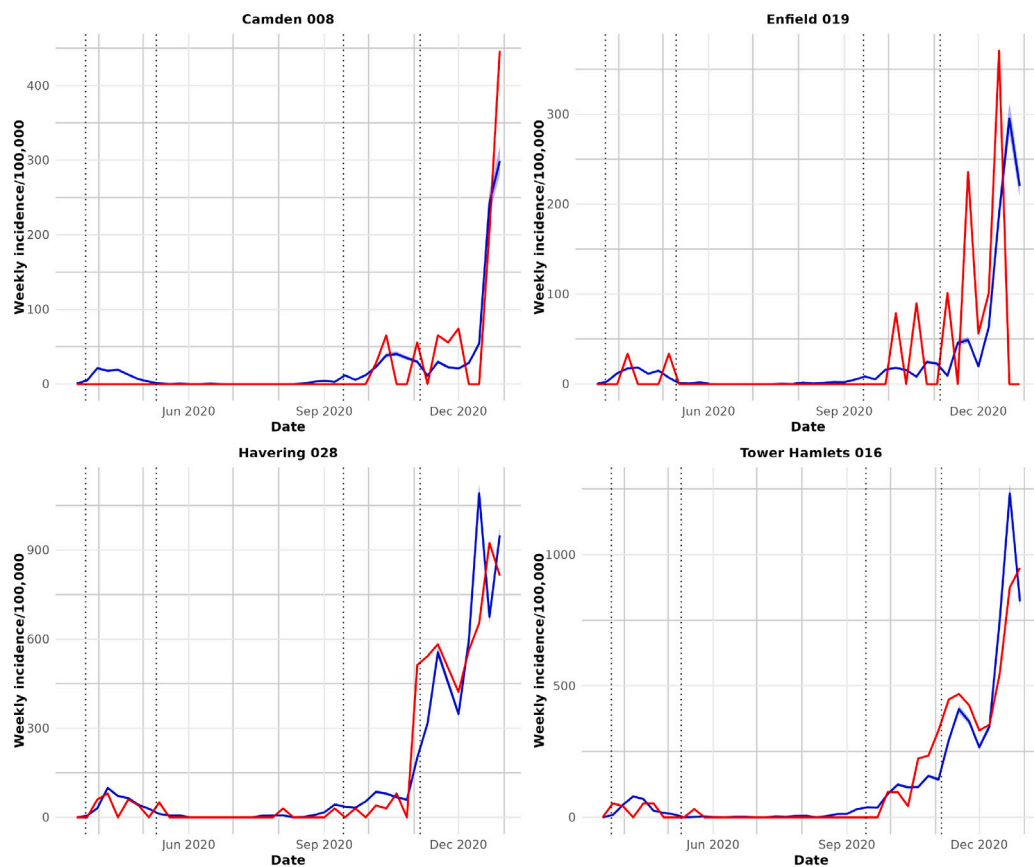


Fig. A.9. Temporal series depicting the observed (in Red color) and fitted (in Blue color with gray colored band) weekly incidence per 100,000 population for four selected MSOAs: Camden 008 (top-left), Enfield 019 (top-right), Havering 028 (bottom-left), and Tower Hamlets 016 (bottom-right).

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